AGRIVIVE / CHAIN-TANK

TELANGANA / CHAIN-TANK INSPIRED / GRAVITY TRANSFER COLLECTOR

SERIES 24 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



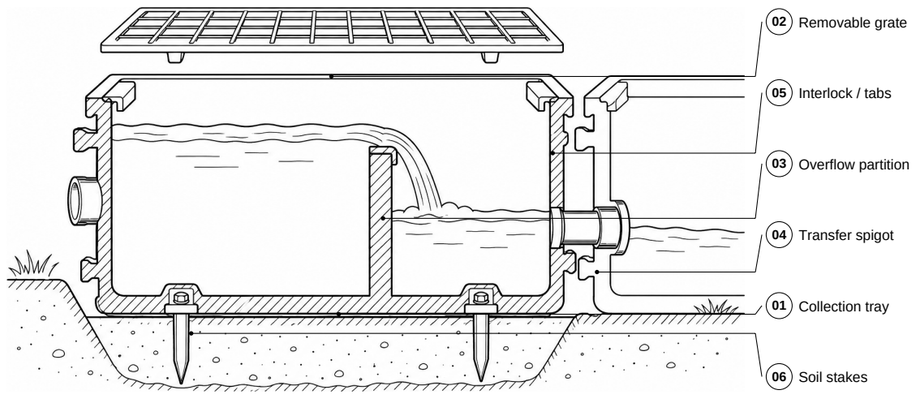
STATE CONTEXT / INDIGENOUS INSPIRATION

Telangana's tank-restoration programmes recognise linked tanks and their catchment flow paths. CHAIN-TANK proposes small linked collectors for a trial gravity route; it does not reproduce a tank network's storage or flood function.

Practice source: Telangana government: Mission Kakatiya general guidelines | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 180 L x 120 W x 100 H

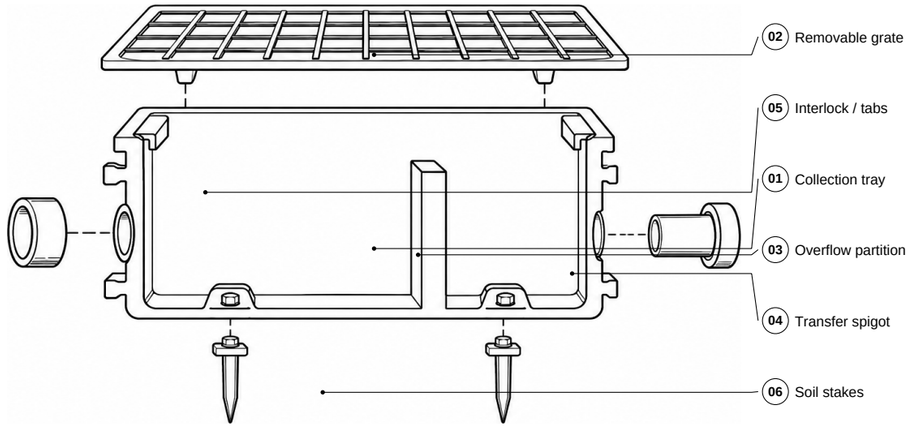


DESIGN / OPERATION NOTES

Seat the first tray level and the next collector downstream. Grate shown lifted for service.
Connect transfer fittings and retain an open overflow path.
Remove grates to clear sediment before rechecking the route.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
180 L x 120 W x 100 H

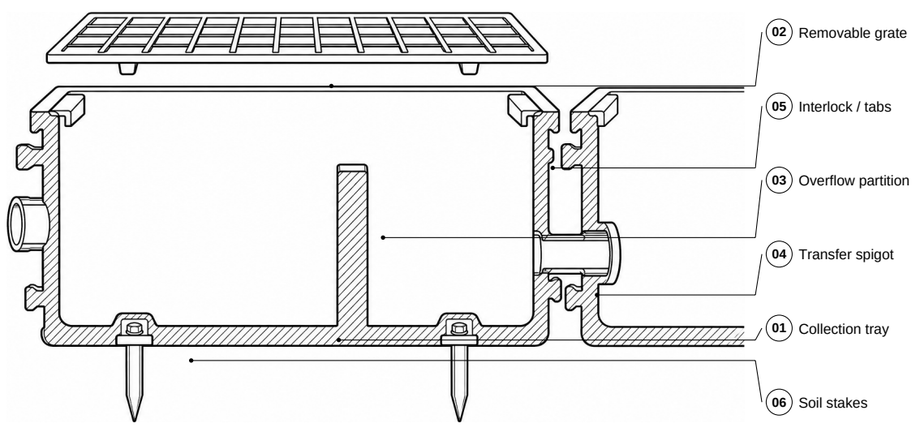


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

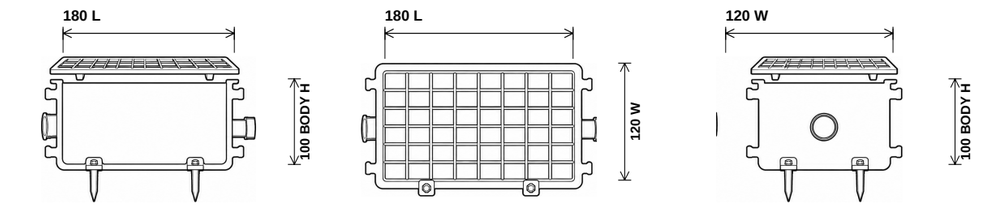
SECTION BASIS / mm
Transfer outlet32 ID; inlet screen and overflow crest TBD



DESIGN / OPERATION NOTES

Connect transfer fittings and retain an open overflow path.
Remove grates to clear sediment before rechecking the route.

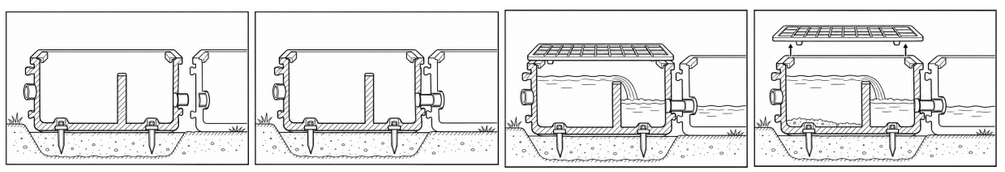
03 ORTHOGRAPHIC STUDIES / mm



FRONT / BODY **TOP / BODY** **END / BODY**

Each body 180 L x 120 W x 100 H; transfer DIA 32 ID. Crest, connection offsets and anchors TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 SET FIRST TRAY **2 LINK DOWNHILL** **3 FIT GRATE** **4 CLEAN / CHECK**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Body L x W x H	**180 x 120 x 100**
Transfer outlet ID	**32**
Overflow crest level	**TBD**

02 GEOMETRIC CHECK

TRANSFER OUTLET AREA

804.2 mm2

pi/4 x 32^2
Must be gravity-linked; does not recreate tank networks

03 PARTS / QTY

01 Collection tray	**1**
02 Removable grate	**1**
03 Overflow partition	**Integral**
04 Transfer spigot	**Integral**
05 Interlock / tabs	**Integral**
06 Soil stakes	**2**

Body/grate: candidate polymer. Gravity couplings and seals: material/fit TBD.

04 INTERFACES / CHECKS

Survey invert levels and available gravity fall.

Test blockage, coupling leakage and overflow bypass.

Check anchorage and service access.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-24 / P01	CHAIN-TANK	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	24 / 28